BRO Academy

THE BUILD

Lesson 6

Lesson Plan

1
Automatization - Why do we need it

2
Bundlers - Types and Features

3
Node.js -> npm -> package.json

4
Vite

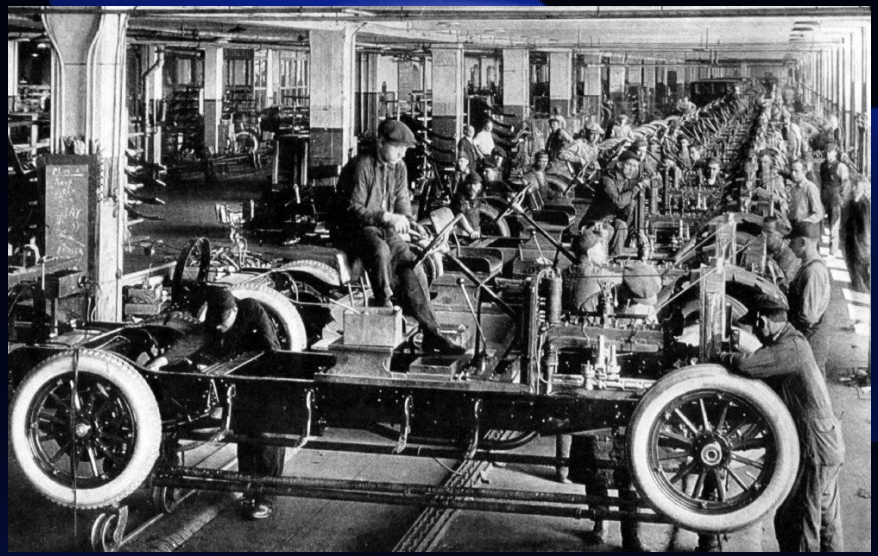
Automatization

is the process of transferring routine development tasks that can be done manually

✅ It **saves the developer's time**

✅ helps to **find and fix errors**

✅ generates code that **saves the browser's resources** and **speeds up website loading**.

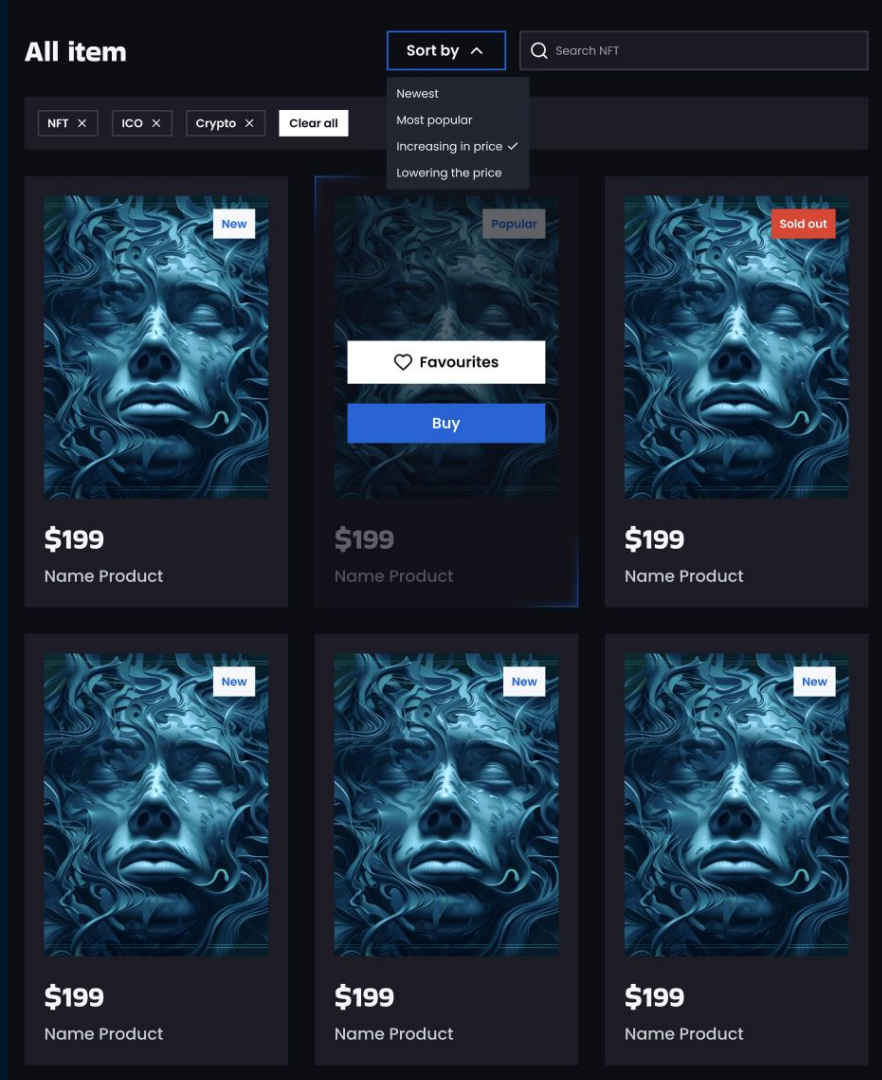


BEFORE

AFTER

Problems

- **Code duplication makes maintenance difficult**
  - wrong copy & paste
  - missing some places
  - simple changes should be done repeatedly across entire project
- **Some files size may slow down a page loading**
  - photos
  - videos
  - scripts, styles



Would be great if we can

- **Extract repeated markup and styles** into a separate components

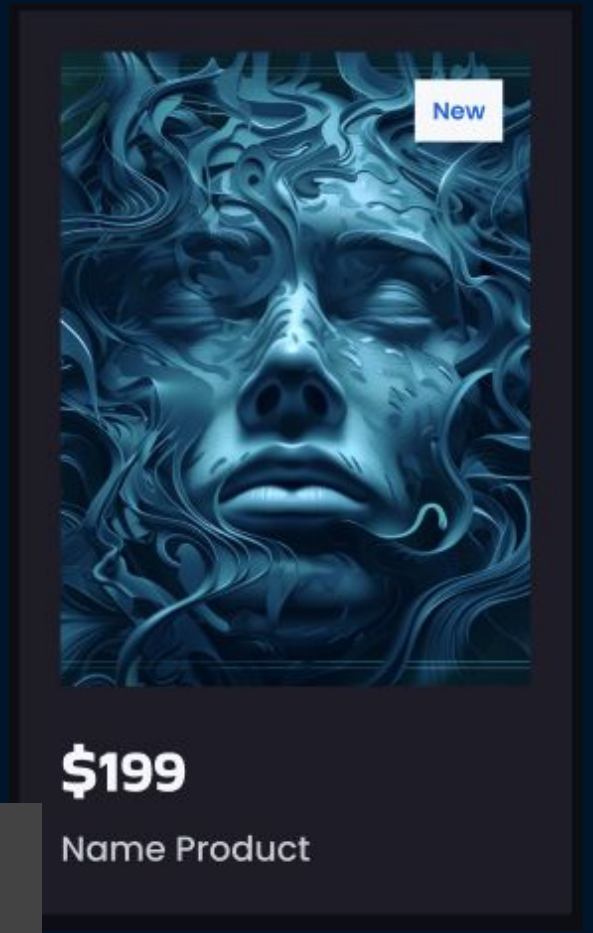
```
{% macro card(title, text, tag) %}

<div class="card">...</div>

{% endmacro %}
```

```
.card {
…
}
```

- Use that component as some short name on the place

```
{% from "card.twig" import card %}

{% card("$199", "Name Product", "New") %}
```

```
@use "card.scss"
```



Popular builders

- **Gulp** – Task runner, can not built without plugins
- **Webpack** – The most popular and powerful, but complex
- **Parcel** – Simple, but slow
- **Vite** – Fast and convenient

💡 **We will use Vite**

PARCEL

Node.js

– JavaScript Runtime
Outside the Browser

📌 **Why do we need it?**
✅ **Allows running tools** including our build
✅ **Manages dependencies** in a project
✅ **Works with packages**

NPM

Node Package Manager is a **package manager for JavaScript**.

📌 **Main functions of npm:**

✅ **Installs and manages libraries and tools**

✅ **Works through the command line (terminal)**

✅ **Uses `package.json` to store project dependencies**

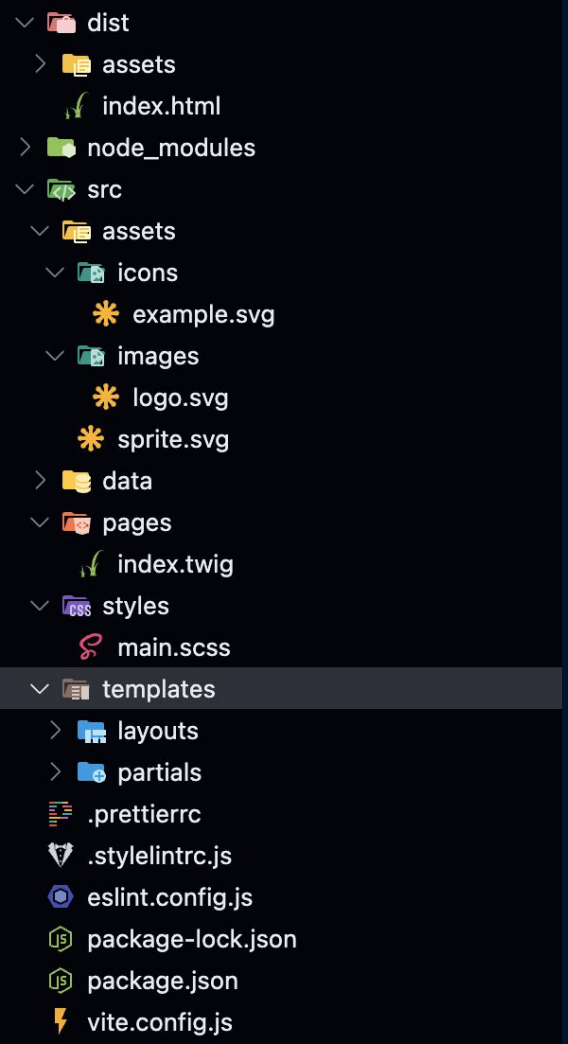
💡 **npm comes bundled with Node.js** – if you've installed Node.js, you already have npm!


Let’s add the Build

Initialize

```
create new branch

cd ..
npx @bro-academy/create-vite-bundler binabox-username

Override? Yes.
Initialize a git repository? No.
```

your binabox project folder



- `dist/` — final build of the project
  - `assets/` — everything imported and not bundled into html
  - `index.html` — compiled HTML file
- `src/` — source files of the project
  - `assets/` — image storage
    - `icons/` — possibly SVG sprites
    - `images/` — regular images
  - `styles/` — source SCSS files
  - `pages/` — Twig page files
  - `templates/` — Twig template files
    - `layouts/` — base wrapper tags that we have on all pages
    - `partials/` — reusable components, e.g., `item-card.twig`

Configuration Files

- `.prettierrc`, `.stylelintrc.js`, `eslint.config.js` — configuration for formatting and linting code.
- `.gitignore` — files excluded from the repository.
- `vite.config.js` — Vite configuration file, responsible for automating the build process.
- `package.json`, `package-lock.json` — list of dependencies and scripts for working with the project.

.gitignore

.DS_Store

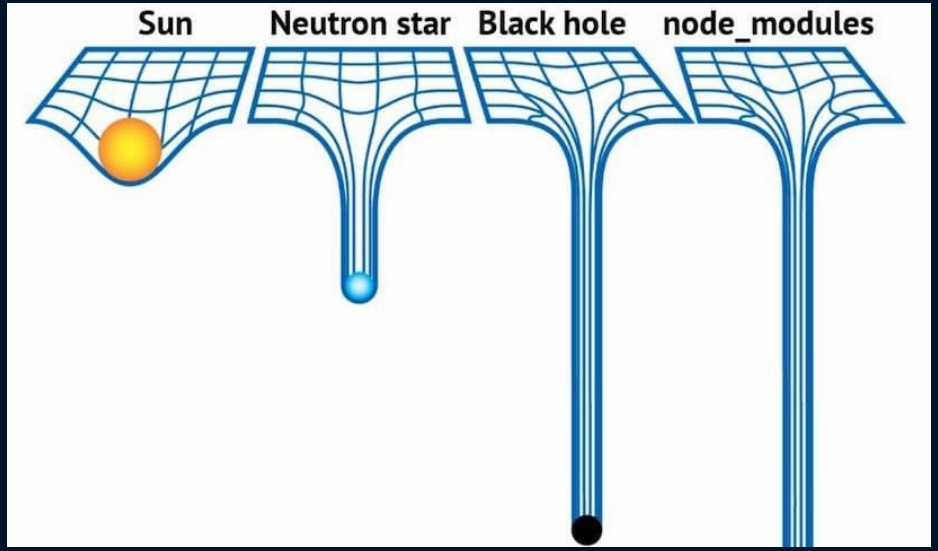
Thumbs.db

***.log**

node_modules/ – too heavy

dist/ – version for prod

.gititnore from the build



Build commands for start

`npm run build` – will generate site (static build)

`npm run dev` – will generate site, watch files and **re-run build on change** (**live server**)

npm run preview - will just create a live server

`ctrl + c` – stop watch

package.json

↯ **The package.json file stores project commands and dependencies.**

Run in the Terminal:

▶️ `npm run build`

```
{
 "name": "binabox",
 "version": "1.0.0",
 "scripts": {
   "dev": "vite dev",
   "build": "vite build",
   "preview": "vite preview",
 },
 "devDependencies": {
   "@vituum/vite-plugin-postcss": "^1.1.0",
   "@vituum/vite-plugin-twig": "^1.1.0",
   "sass-embedded": "^1.88.0",
   "vite": "^6.3.5",
   "vituum": "^1.2.0"
 }
}
```

Volta

Volta is a **JavaScript toolchain manager** that helps developers manage Node.js runtimes and package managers (like npm, Yarn, and pnpm) in a fast and consistent way.

We have pre-installed volta for you as a part of environment setup

When it is installed you may forget about it

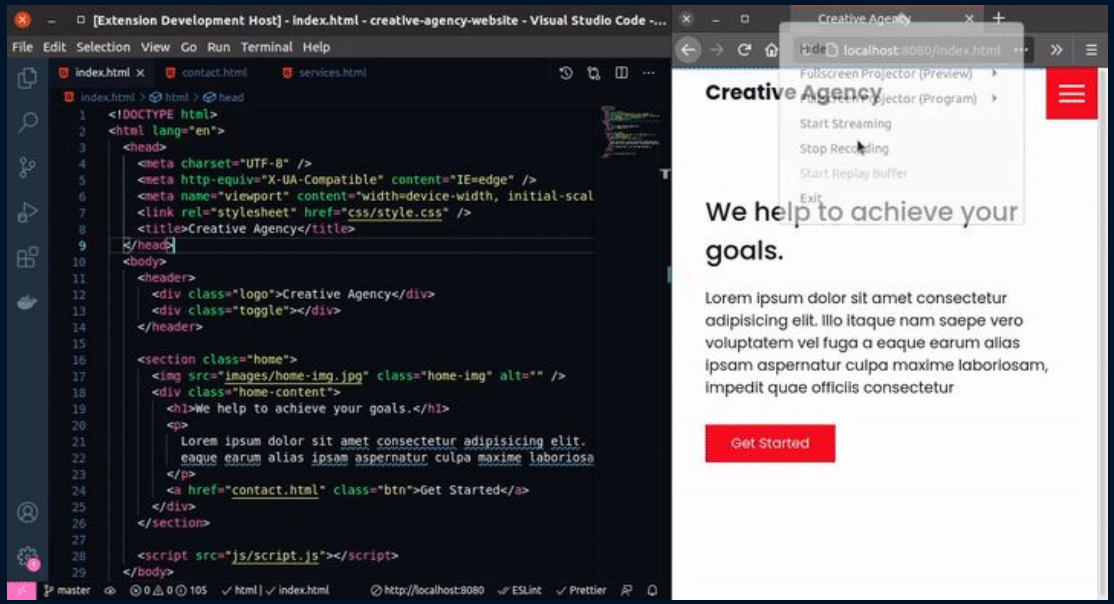
Live Server

Automatic Page Refresh

Edit →

Save →

Browser Updates!



CSS Processing

✅ Allows component approach

```scss
// SCSS

$white: #fff;
$black: #000;

.button {
    text-align: center;
    color: $white;
    background-color: $black;
    border: 1px solid $black;
    padding: 1em 4em;
    &:hover {
        color: $black;
        background-color: $white;
        transition: 0.15s all linear;
    }
}
```

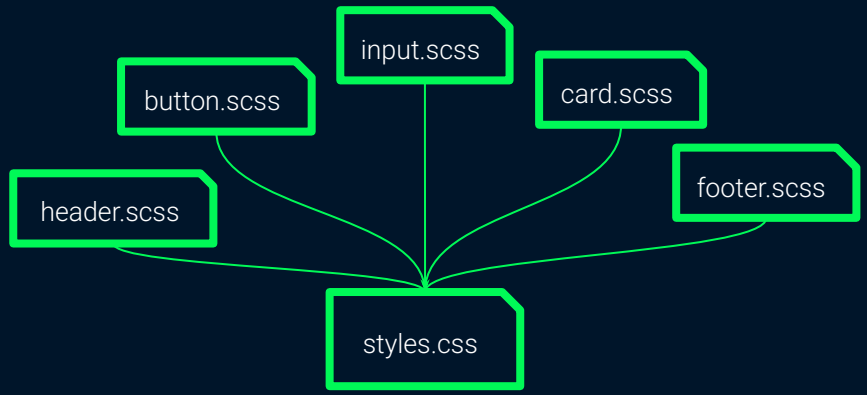
```css
/* CSS */

.button {
    text-align: center;
    color: #fff;
    background-color: #000;
    border: 1px solid #000;
    padding: 1em 4em;
}

.button:hover {
    color:#000;
    background-color: #fff;
    transition: 0.15s all linear;
}
```

✅ Make styles more cross browser

```
div {
    transition: all
}
```

```
1  div {
2    -webkit-transition: all 1s ease-out;
3    -o-transition: all 1s ease-out;
4    transition: all 1s ease-out;
5  }
```

✅ Combines files



✅ Minifies code

```
.entry-content p {
font-size: 14px !important;
}

.entry-content ul li {
font-size: 14px !important;
}

.product_item p a {
 color: #000;
 padding: 10px 0px 0px 0;
 margin-bottom: 5px;
```

→

```
.entry-content p,.entry-content ul li
{font-size:14px!important}.product_item
p a{color:#000;padding:10px 0 0;margin-
bottom:5px;border-bottom:none}
```

Source maps

```
#topmenu #desktop-menu          desktopmenu.scss:4
ul li {
    float: left;
    margin: ▸ 20px 20px 15px 20px;
    padding-bottom: 5px;
    line-height: 1;
    position: relative;
    cursor: pointer;
    text-align: center;
}
```

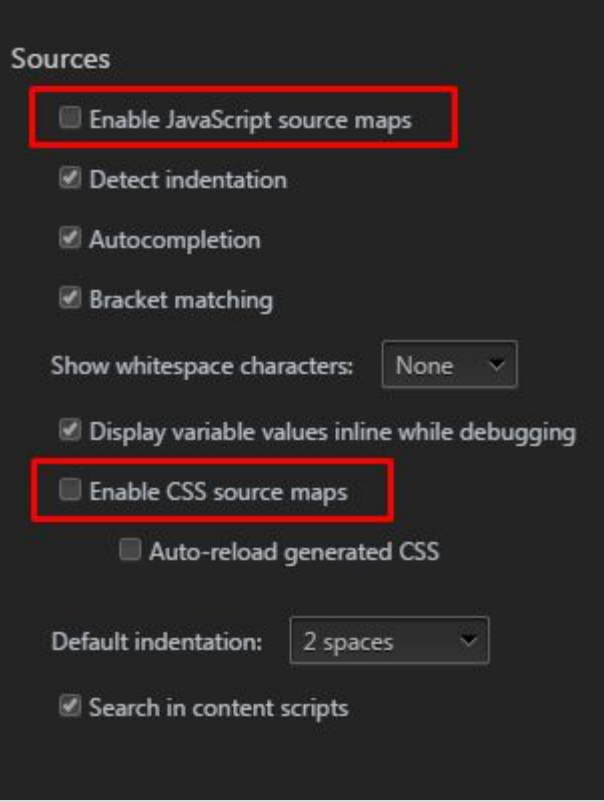


Image Optimization & Responsive Images

📌 **Processes images for performance optimization:**

💡 **Why?**
✅ Converts images to **different sizes for responsive design**.
✅ Optimizes file size without losing quality.
✅ Generates modern formats like **WebP and AVIF**.

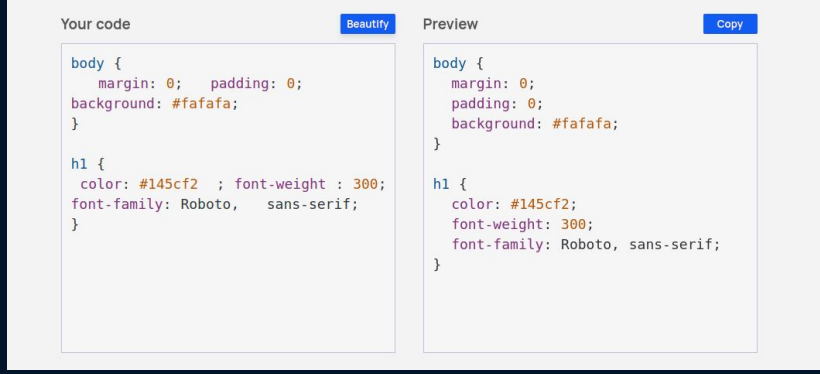
📌 **Combines multiple SVG icons into a single optimized sprite file:**

- Creates **SVG sprites** to reduce HTTP requests.

💡 **We will use it later**


Before
Quality: great
78
PageSpeed Score
Original image
JPG
1.2 MB
Optimize
After
Quality: great
100
PageSpeed Score
Optimized image
WebP
0.3 MB

prettier



What does it do?

- Automatically **formats code** to follow a consistent style.
- Works with **CSS, SCSS, JavaScript, JSON, and more**.
- **Fixes indentation, line breaks, spacing, and quotes**.

💡 **Why use prettier?**

✅ **Formats code automatically** so you don’t have to.

✅ **Works with multiple languages** (JavaScript, CSS, JSON, etc.).

✅ **Prevents unnecessary code diffs** in Git (e.g., inconsistent spacing).

stylelint, eslint + htmllint

📌 **Ensures consistent coding style** 🎨

✅ **Checks CSS/SCSS and HTML** code for errors and enforces coding style rules.

✅ **Prevents common mistakes** (e.g., unknown properties, missing quotes).

✅ **Ensures the use of BEM** naming convention for class names.

🔹 **Keeps your styles clean and maintainable!** 🚀

```
> npx stylelint '**/*.scss'


src/styles/main.scss
  60:1   ✖  ".ololo--ssss__sss" selector should be written by the BEM naming methodology:    selector-class-pattern
            https://en.bem.info/methodology/naming-convention/#alternative-naming-schemes
  61:15  ✖  Unexpected named color "red"                                                     color-named

✖ 2 problems (2 errors, 0 warnings)
```

📌 The Key Differences Between These Tools

Feature	`stylelint`	`eslint`	`prettier`
Controls formatting in the editor	Able to setup	HTML & JS	✅ Yes
Linting & error checking	✅ Yes	✅ Yes	❌ No
Automatic code formatting	Able to setup for css/scss	Able to setup	✅ Yes
Works with	**CSS & SCSS**	**HTML & JS**	**JS, CSS, JSON, etc.**
Example rules	Class naming, color rules		Line breaks, spacing

Build commands for start

npm run lint - checks your HTML for mistakes

npm run lint:fix - will fix most popular of them

npm run stylelint - checks your CSS/SCSS for mistakes

npm run stylelint:fix - will fix most popular of them

npm run format - will fix code style things

npm run format:check - will fix code style things

npm run format:lint - will fix code style things and then most popular mistakes

`ctrl + c` – stop watch

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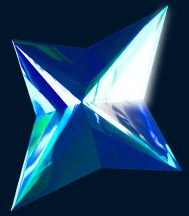


Break time

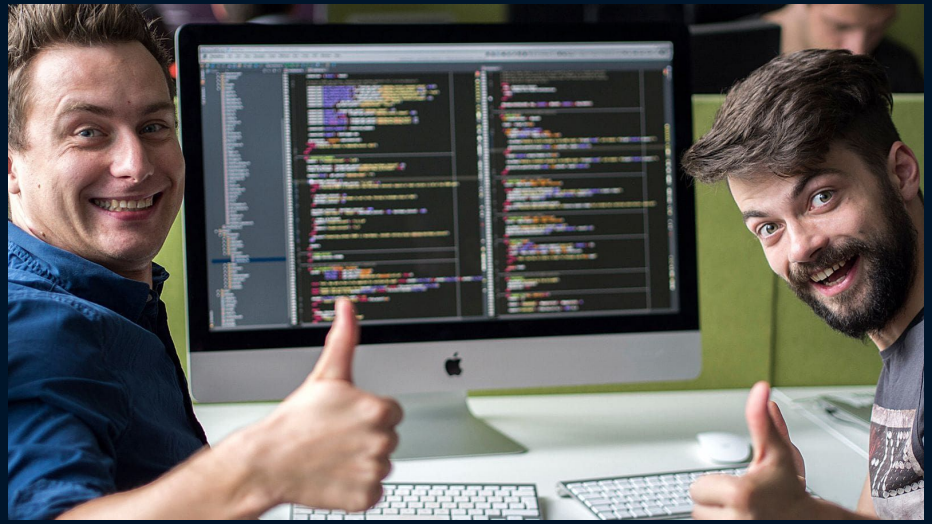
Summary – Now We Know:

✅ What a bundler is and why we need it

✅ What is Vite

✅ How to install and use them

✅ How to automate work with your project



Code easier, faster, and more efficiently 🚀

Quality Criteria for HTML Course

❤️ Mandatory for passing the course

💛 Required for the highest grade

💚 Optional

❤️ The build is used

❤️ Folders `dist` and `node_modules` should not be in the GitHub repository

❤️ The project builds correctly

❤️ No red errors in the console during the build process

💛 No warnings in the console during the build process

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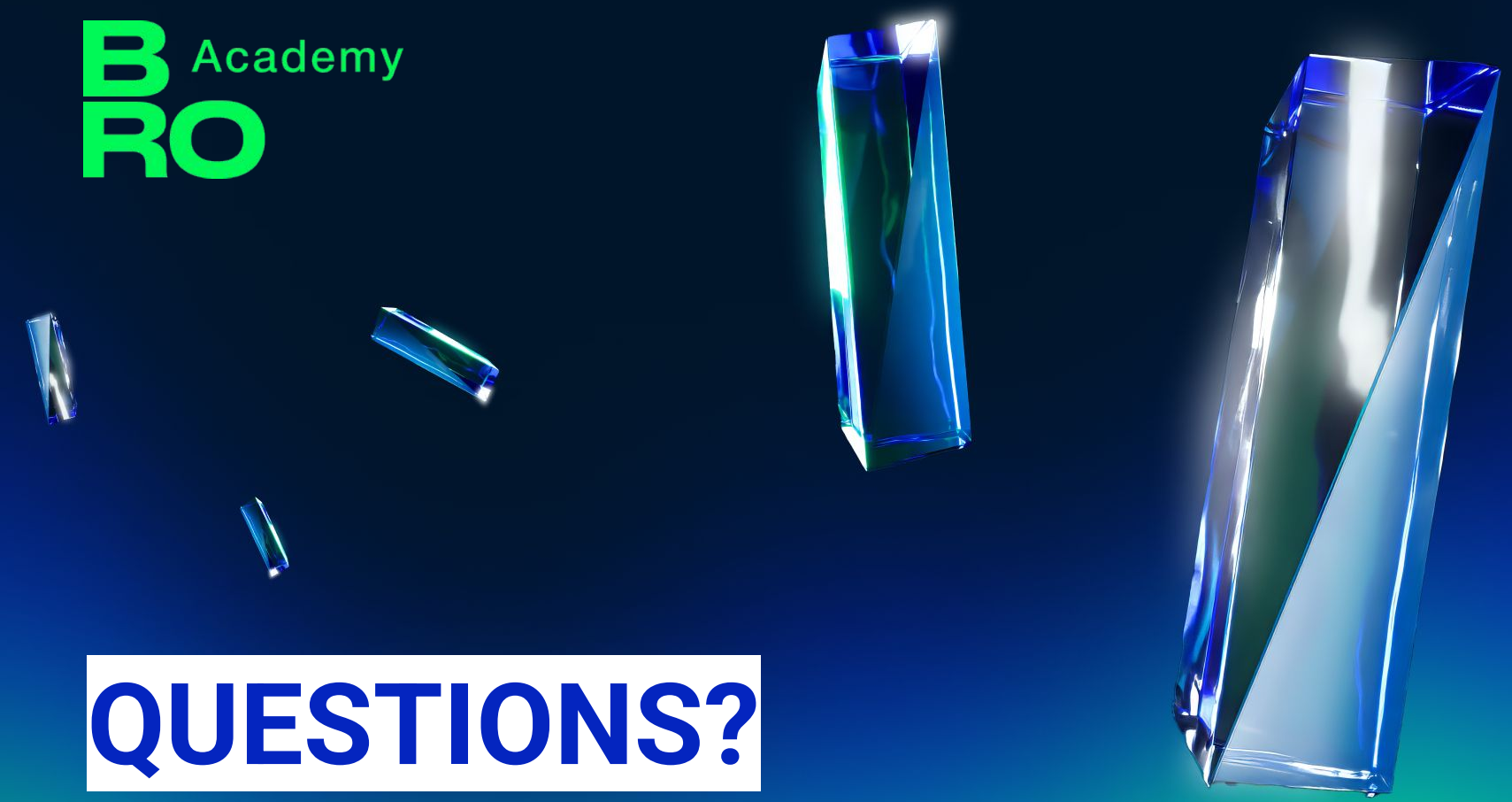
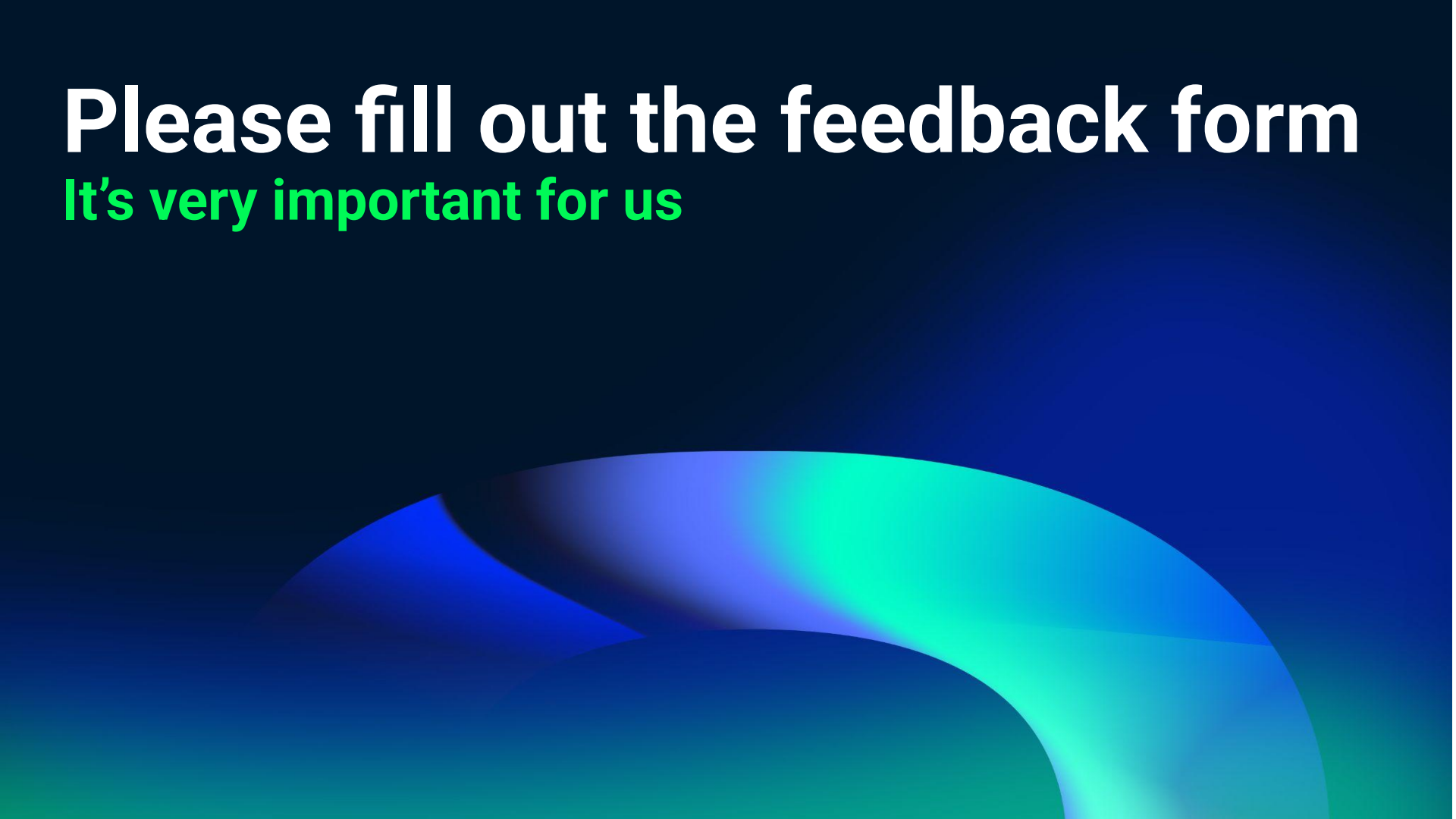


QUESTIONS?

Please fill out the feedback form

It’s very important for us




THANK YOU!
Have a good evening!